Amaris Huang

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EDUCATION

Duke University, Pratt School of Engineering **Durham, NC**
Biomedical Engineering (B.S. E.), Computer Science (B.A.) *May 2025*

Honors: Graduation with Distinction (2025); National Academy of Engineering Grand Challenges Scholar (2025); Pratt Research Fellows (2023); The Jackson Laboratory National Cancer Institute Fellow (2022)

WORK EXPERIENCE

Beghou Consulting **New York City, NY**
Associate Consultant *Aug 2026 - Present*
- Built a HIPAA-compliant data pipeline (AWS, Databricks) for patient access and value assessment in a national oncology drug launch
- Worked on a field effectiveness reporting engagement for a large pharmaceutical client to inform commercial resource allocation

MIT **Cambridge, MA**
Research Associate *Aug 2025 - July 2026*
- Addressed a gap in commercially viable diagnostics for Rocio and Powassan virus by leading 0-to-1 development and product iteration of two accessible, high-sensitivity lateral flow tests from concept through clinical validation
- Enabled 2 publications by translating diagnostic data into technical presentations for academic and industry collaborators

North Carolina Integrated Care for Kids **Durham, NC**
Health Policy Intern *Sep 2022 - May 2025*
- Conducted 25+ physician interviews to identify unmet needs and financing mechanisms for scaling food insecurity programs nationwide, informing North Carolina Medicaid's transformation strategy and an Academic Pediatrics publication.
- Co-first-authored a drafted manuscript on novel implementation science and human-centered design framework to co-design a scalable, multi-sector workflow for complex care systems, using food insecurity as a use case

Duke University **Durham, NC**
Pratt Research Fellow, Chilkoti Group *Feb 2023 - May 2025*
- Led cross-functional development of point-of-care diagnostics for Ebola and Ricin, translating protein engineering, AlphaFold, and thermodynamic modeling into microfluidic product design; communicated results to diverse stakeholders at a national conference
- Cut evaluation time 70% by building an automated Python analysis pipeline, streamlining testing across 100+ prototype iterations to deliver a Department of Defense-funded prototype; awarded over $13,000 in personal competitive grants and fellowships

PROJECT EXPERIENCE

Deep Learning for Biology | *Python, PyTorch, TensorFlow, sci-lit learn*
- Built a generative model predicting protein sequences from transcription factor DNA-binding motifs by fine-tuning DNABert and T5 transformer architectures on a curated dataset of 26,000 pairs, achieving support vector machine test accuracy of ~ 82.5%
- Implemented CNNs and Vision Transformers to classify Cutaneous T-Cell Lymphoma histopathology images with ~ 98% accuracy

Senior Design Capstone | *Project Management, Computer-Aided Design (CAD)*
- Shipped an ergonomic gynecology table for an underserved Peruvian hospital through iterative CAD-driven design cycles, delivering a validated solution within a $200 budget constraint and advancing it to field testing; won the capstone design competition

LEADERSHIP EXPERIENCE

Duke-Ignite, *Program Mentor and Bass Connections Researcher* *Sep 2022 - May 2025*
- Launched a study evaluating program impact for 150+ participants by securing IRB approval and designing evaluation metrics
- Advanced 10 student-led prototypes from concept to minimum viable product (MVP) by teaching iterative human-centered design
- Presented findings at two national engineering conferences and two university research symposia, and published in ASEE

Kode with Klossy, *Teaching Assistant and Workshop Facilitator* *Oct 2021 - Jun 2025*
- Taught front-end fundamentals to 100+ students in an international coding camp by leading instruction in HTML/CSS/JavaScript and user-oriented design, guiding students through building an entrepreneurial passion project from concept to minimum viable product
- Launched a self-directed outreach workshop to engage the local community by recruiting participants and designing the curriculum

SKILLS AND INTERESTS

Coding Skills: Python, SQL, R, Cloud-Based Platforms (AWS, Databricks), Git
Product Development Skills: Jira, Figma, Adobe Creative Suite, CAD, Arduino IDE, Circuit Design, Laser-Cutting
Languages: Mandarin Chinese (Native Fluency), English (Native Fluency), Spanish (Intermediate Fluency)